Module 6: Introduction



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Module Overview

This foundational module introduces students to the field of artificial intelligence, tracing its evolution from early philosophical questions about machine intelligence through the Dartmouth Conference to modern breakthroughs. Students will explore different definitions and paradigms of AI, including symbolic AI, connectionism, and hybrid approaches, while understanding the distinction between narrow AI, general AI, and superintelligence. The module covers key concepts like the Turing Test, intelligent agents, and the AI effect, providing students with a conceptual framework for understanding subsequent technical content.

Learning Objectives

By the end of this module, you should be able to:

- Define artificial intelligence and distinguish between different types of AI systems
- Trace the historical development of AI and identify key milestones and pioneers
- Explain fundamental AI concepts, including agents, environments, and rationality
- Analyze the philosophical foundations and debates surrounding machine intelligence
- Evaluate current AI capabilities and limitations

Task List

- Participate in the weekly discussion board
- Solve and submit the weekly assignment
- Solve module 6 Quiz
- Additional reading:
  - 1. Badhan Chandra Das, M. Hadi Amini, and Yanzhao Wu. 2025. Security and Privacy Challenges of Large Language Models: A Survey. ACM Comput. Surv. 57, 6, Article 152 (June 2025), 39 pages. **https://doi.org/10.1145/3712001 (https://doi.org/10.1145/3712001)**
  - Devang Chavan (2024). Complete Guide to Text Preprocessing in NLP. **https://medium.com/@devangchavan0204/complete-guide-to-text-preprocessing-in-nlp-b4092c104d3e (https://medium.com/@devangchavan0204/complete-guide-to-text-preprocessing-in-nlp-b4092c104d3e)**
  - Ahmad Mizan Nur Haq (2023). Exploring NLP Techniques for Text Summarization. **https://medium.com/data-and-beyond/exploring-nlp-techniques-for-text-summarization-8038426e13de (https://medium.com/data-and-beyond/exploring-nlp-techniques-for-text-summarization-8038426e13de)**

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- Mayur Koshti (2025). Build Your Own AI ChatBot in Python — Step-by-Step with Code. https://medium.com/python-in-plain-english/build-your-own-ai-chatbot-in-python-step-by-step-with-code-d7728b927374 (https://medium.com/python-in-plain-english/build-your-own-ai-chatbot-in-python-step-by-step-with-code-d7728b927374)
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- GeeksforGeeks. (2023). Ethical Considerations in Natural Language Processing. https://www.geeksforgeeks.org/nlp/ethical-considerations-in-natural-language-processing-bias-fairness-and-privacy/ (https://www.geeksforgeeks.org/nlp/ethical-considerations-in-natural-language-processing-bias-fairness-and-privacy/)
- GeeksforGeeks. (2025). Named Entity Recognition. https://www.geeksforgeeks.org/nlp/named-entity-recognition/ (https://www.geeksforgeeks.org/nlp/named-entity-recognition/)
- GeeksforGeeks. (2022). Part of Speech Tagging with NLTK. https://www.geeksforgeeks.org/python/part-speech-tagging-stop-

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